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# **basic education**

Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

## **SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS**

**AGRICULTURAL SCIENCES P1**

**MAY/JUNE 2024**

**MARKING GUIDELINES**

**MARKS: 150**

**These marking guidelines consist of 12 pages.**

**SECTION A****QUESTION 1**

|     |        |                                                          |          |      |
|-----|--------|----------------------------------------------------------|----------|------|
| 1.1 | 1.1.1  | B ✓✓                                                     |          |      |
|     | 1.1.2  | D ✓✓                                                     |          |      |
|     | 1.1.3  | A ✓✓                                                     |          |      |
|     | 1.1.4  | C ✓✓                                                     |          |      |
|     | 1.1.5  | B ✓✓                                                     |          |      |
|     | 1.1.6  | C ✓✓                                                     |          |      |
|     | 1.1.7  | D ✓✓                                                     |          |      |
|     | 1.1.8  | A ✓✓                                                     |          |      |
|     | 1.1.9  | B ✓✓                                                     |          |      |
|     | 1.1.10 | C ✓✓                                                     | (10 x 2) | (20) |
| 1.2 | 1.2.1  | None ✓✓                                                  |          |      |
|     | 1.2.2  | Both A and B ✓✓                                          |          |      |
|     | 1.2.3  | B only ✓✓                                                |          |      |
|     | 1.2.4  | B only ✓✓                                                |          |      |
|     | 1.2.5  | A only ✓✓                                                | (5 x 2)  | (10) |
| 1.3 | 1.3.1  | Retro/reverse peristalsis/regurgitation ✓✓               |          |      |
|     | 1.3.2  | Sustainable medication/integrated diseases management ✓✓ |          |      |
|     | 1.3.3  | Ovigenesis/oogenesis ✓✓                                  |          |      |
|     | 1.3.4  | Repeat breeders ✓✓                                       |          |      |
|     | 1.3.5  | Impotence ✓✓                                             | (5 x 2)  | (10) |
| 1.4 | 1.4.1  | Diffusion/passive absorption ✓                           |          |      |
|     | 1.4.2  | Plywood board ✓                                          |          |      |
|     | 1.4.3  | Oestrus ✓                                                |          |      |
|     | 1.4.4  | Foetal ✓                                                 |          |      |
|     | 1.4.5  | Pistolette/insemination gun ✓                            | (5 x 1)  | (5)  |

**TOTAL SECTION A: 45**

**SECTION B****QUESTION 2: ANIMAL NUTRITION****2.1 The alimentary canal of a farm animal****2.1.1 Classification of the farm animal**

Non-ruminant/monogastric ✓ (1)

**2.1.2 Name of the farm animal**

Fowl/poultry/chicken ✓ (1)

**2.1.3 Reason based on the caeca**

There are two caeca present ✓ (1)

**2.1.4 Identification of the parts****A** Small intestine/duodenum/jejunum/ileum ✓ (1)**C** Rectum/colon ✓ (1)**D** Cloaca/vent ✓ (1)**2.2 Micro-organisms in the reticulo-rumen****2.2.1 TWO examples of micro-organisms found in the reticulo-rumen**

- Bacteria ✓
  - Protozoa ✓
  - Fungi ✓
  - Archaea ✓
  - Viruses ✓
- (Any 2) (2)

**2.2.2 TWO requirements for the normal functioning of micro-organisms**

- Anaerobic conditions ✓
  - The presence of carbon dioxide ✓
  - A pH of 5,5 - 6,5/slightly acidic ✓
  - A temperature of 38 - 42 °C/warm environment ✓
  - Regular intake of feed ✓
  - Removal of the waste products ✓
  - Osmotic conditions/presence of moisture ✓
  - The presence of volatile fatty acids ✓
  - The presence of sufficient nutrients ✓
- (Any 2) (2)

**2.3 The schematic representation of the types of feed****2.3.1 Identification of the types of feed**

- A** Concentrates ✓ (1)  
**B** Roughages ✓ (1)

**2.3.2 Identification of letter**

- (a)** Succulent feed - C ✓ (1)  
**(b)** Maize meal - D ✓ (1)  
**(c)** Increases the bulkiness of the ration - B/C ✓ (1)

**2.4 Digestibility co-efficient****2.4.1 Calculation of the digestibility co-efficient of the hay**

$$\text{DMI (kg)} = \frac{90}{100} \times 18 \text{ kg} = 16,2 \text{ kg} \checkmark$$

$$\text{DC} = \frac{\text{Dry matter intake (kg)} - \text{dry mass of manure (kg)}}{\text{Dry matter intake (kg)}} \times 100 \checkmark$$

$$= \frac{16,2 \text{ kg} - 5 \text{ kg}}{16,2 \text{ kg}} \times 100 \checkmark$$

$$= 69,1 \checkmark \% \checkmark \quad (5)$$

**2.4.2 Implication**

- 69,1% ✓ of the feed was digested and absorbed ✓
- 30,9% ✓ was excreted ✓
- Highly ✓ digested and absorbed ✓ (Any 1) (2)

**2.5 Re-arrangement of the sources of protein**

- Eggs ✓ (1)
- Milk ✓ (1)
- Lucerne ✓ (1)

**2.6 Nutritive Ratio****2.6.1 Definition of the nutritive ratio**

NR is the ratio between the digestible protein ✓ and digestible non-nitrogen compounds in a ration or a feed ✓ (2)

**2.6.2 Calculation of the nutritive ratio**

$$\text{NR} = 1 : \frac{\% \text{TDN} - \% \text{DP}}{\% \text{DP}} \checkmark$$

$$= 1 : \frac{79\% - 7\%}{7\%} \checkmark$$

$$\text{NR} = 1:10 \checkmark$$

**OR**

$$\text{NR} = 1 : \frac{\% \text{ digestible non-nitrogen nutrients}}{\% \text{ Digestible protein}} \checkmark$$

$$= 1 : \frac{72\%}{7\%} \checkmark$$

$$\text{NR} = 1:10 \checkmark$$

(3)

**2.7 Fodder flow****2.7.1 Identification of month with lowest feed available**July  $\checkmark$ 

(1)

**2.7.2 REASON to support the decline in quantities**Adverse environmental conditions  $\checkmark$ 

(1)

**2.7.3 Calculation of the total feed required from January - June**

- 50+50+60+60+70+60 tons  $\checkmark$
- 350 tons  $\checkmark$

(2)

**2.7.4 The trend of feed available from August - December**There is an increase in the feed available from August - December  $\checkmark$ 

(1)

**[35]**

**QUESTION 3: ANIMAL PRODUCTION****3.1 Farming and production systems****3.1.1 Identification of farming and production systems**

- (a) **Farming system** - Subsistence ✓ (1)  
 (b) **Production system** - Extensive ✓ (1)

**3.1.2 Justification for extensive production system**

- The animals search food for themselves ✓
- There are few animals in large area of land ✓ (Any 1) (1)

**3.1.3 Differentiate between subsistence and commercial farming**

**Subsistence farming** - Practiced mainly to sustain the family/  
 less impact to the environment/capital input/labour/technology ✓ (1)

**Commercial farming** - Practiced mainly for selling/generate profit/  
 more impact to the environment/capital input/labour/technology ✓ (1)

**3.1.4 TWO factors to increase animal production**

- Nutrition/Feeding ✓
- Breeding/reproduction ✓
- General management ✓
- Environment ✓ (Any 2) (2)

**3.2 The equipment used in different intensive production systems****3.2.1 Identification of the equipment**

- A Nipple drinkers ✓ (1)  
 B Feeding trough/feeder ✓ (1)  
 C Infra-red light ✓ (1)

**3.2.2 TWO reasons for keeping chickens in a shelter**

- Protection against extreme weather conditions ✓
- Protection from predators/theft ✓
- For easy management/handling/control ✓
- Optimum production ✓
- Upholding biosecurity ✓ (Any 2) (2)

### 3.3 Animal handling

#### 3.3.1 Reasons for handling cattle

- A Dehorning ✓ (1)  
B Milking ✓ (1)

#### 3.3.2 TWO basic guidelines for handling cattle

- Announce your approach through a touch to the animals ✓
- Avoid the fight zone ✓
- Use proper handling facilities ✓
- Always leave yourselves a way to get out ✓
- Never prod animals if there is no way to go ✓
- Give them time to adjust before working with them/  
move animals slowly at their own pace ✓
- Avoid entering a small area enclosed with animals ✓
- Avoid shouting ✓
- Avoid whipping/throwing objects at animals ✓
- Work with calves in an area isolated from their mothers ✓
- People handling farm animals should be well trained/skilled/  
knowledgeable ✓ (Any 2) (2)

#### 3.3.3 TWO effects of incorrect handling of farm animals

- Farm animals become more difficult to handle ✓
- Bruises/decrease in quality of products ✓
- Decrease in production/farm animals may die ✓
- Drop in conception rate during AI ✓
- Aggressive/danger to handlers/other animals ✓ (Any 2) (2)

### 3.4 Animal diseases

#### 3.4.1 Completing the missing information

- A Ringworm ✓ (1)  
B Protozoa ✓ (1)  
C Mastitis ✓ (1)  
D Virus ✓ (1)  
E
  - Aggression/excitement/frequent barking/biting everything ✓
  - Froth in the mouth/excessive salivation/drooling ✓
  - Running around/circling ✓
  - Paralysis of lower jaw and tongue/incoordination ✓
  - Seizures and death ✓ (Any 1) (1)

#### 3.4.2 TWO economic implications of contagious animal diseases

- Banning of exports and imports/decrease in international trade ✓
- Decreased production/death ✓
- High treatment/vaccination costs to control/prevent diseases ✓
- Loss of income/profit ✓ (Any 2) (2)

**3.5 Life cycle of a parasite****3.5.1 Classification of the parasite**

Internal/endo parasite ✓

(1)

**3.5.2 Identification of two hosts**

• Sheep ✓

(1)

• Snail ✓

(1)

**3.5.3 TWO pasture management practices to control the parasite**

• Rotational grazing

• Resting the infested pasture ✓

• Avoiding grazing wet areas ✓

• Allow resistant species to graze on the pasture ✓

• Practice zero grazing ✓

• Grazing on clean pastures ✓

• Veld burning/eradication of snails ✓

(Any 2)

(2)

**3.6 External parasites****3.6.1 Identification of parasites****PARASITE A** - Nasal worm ✓

(1)

**PARASITE B** - Mites ✓

(1)

**3.6.2 ONE symptom of nasal worm**

• Sneezing and nasal irritation ✓

• Shaking of head to get rid of the parasite ✓

• Severe thick yellow nasal discharge ✓

• Respiratory distress/difficulty breathing ✓

(Any 1)

(1)

**3.6.3 TWO ways to prevent blowfly strike**

• Correct timing of shearing ✓

• Clipping of soiled wool/crutching ✓

• Cleaning/treatment of wounds ✓

• Tail docking ✓

• Lambing time after shearing ✓

• Breeding and selection of resistant animals ✓

• Waste management/hygiene/sanitation ✓

(Any 2)

(2)

**[35]**

**QUESTION 4: ANIMAL REPRODUCTION****4.1 Male and female reproductive organs of cattle**

- 4.1.1 C ✓ (1)
- 4.1.2 D ✓ (1)
- 4.1.3 A ✓ (1)
- 4.1.4 E ✓ (1)

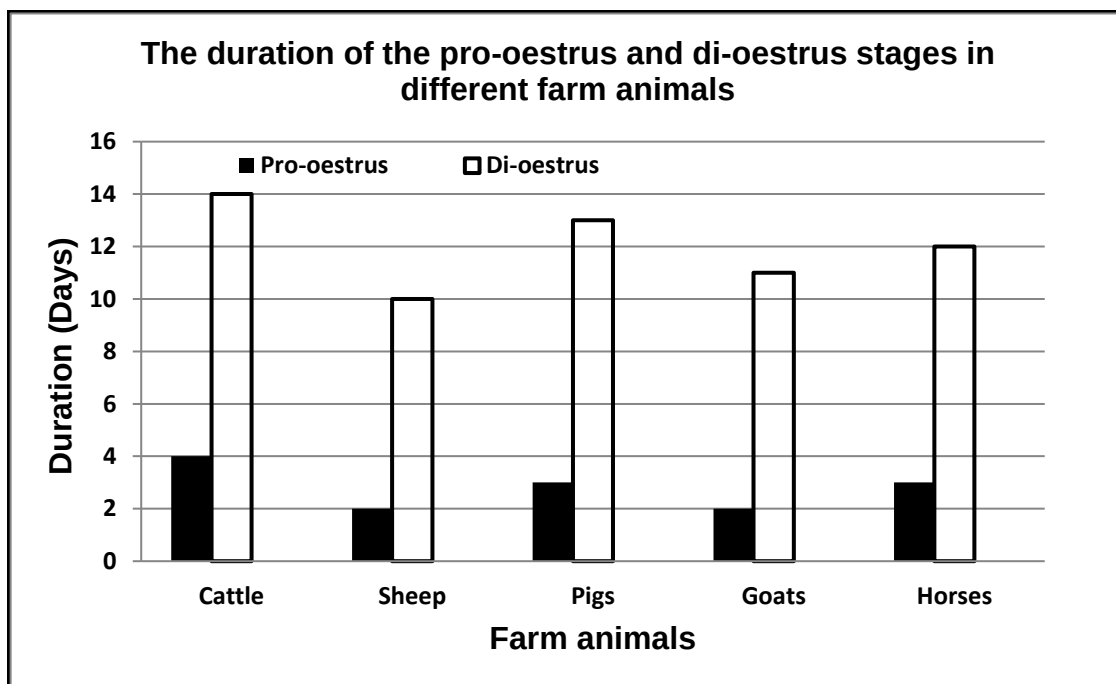
**4.2 Name of the hormones**

- 4.2.1 Follicle stimulating hormone/FSH ✓ (1)
- 4.2.2 Oestrogen ✓ (1)
- 4.2.3 Progesterone ✓ (1)
- 4.2.4 Follicle stimulating hormone/FSH ✓ (1)

**4.3 Sterility and infertility in farm animals**

- 4.3.1 **Name of the conditions leading to sterility in farm animals**  
**Underdeveloped testis or ovaries - Hypoplasia ✓** (1)
- 4.3.2 **Testis stay attached to the body cavity and do not move down to the scrotum - Cryptorchidism ✓** (1)

4.4 **Combined bar graph representing the duration of the pro-oestrus and di-oestrus stages in different farm animals**



**CRITERIA/RUBRIC/MARKING GUIDELINES**

- Correct heading ✓
- X-axis: Correct calibration and labelled (Farm animals) ✓
- Y-axis: Correct calibration and labelled (Duration) ✓
- Correct units (days) ✓
- Combined bar graph ✓
- Accuracy (80%+ correctly plotted) ✓ (6)

4.5 **Synchronization of oestrus**

4.5.1 **Name of the process**

Synchronization of oestrus ✓ (1)

4.5.2 **TWO methods used when performing oestrus synchronization**

- Injection of prostaglandin ✓
- Injection of synthetic progesterone ✓
- Injection of CoSynch oestrus synchronization/GnRH ✓
- MGA/PG mixed into the feed ✓
- Vaginal insertion of hormones/CIDR ✓
- Ear implantations of progesterone ✓ (Any 2) (2)

**4.5.3 TWO disadvantages of the synchronization of oestrus**

- Labour intensive ✓
  - Involves skilled management ✓
  - Expensive/high technology ✓
  - Requires regular pregnancy testing ✓
  - Needs good handling facilities ✓
  - Poor body conditions/nutrition/health will affect the process negatively ✓
- (Any 2) (2)

**4.6 Artificial insemination****4.6.1 Identification of the reproductive technique**

Artificial insemination/AI ✓ (1)

**4.6.2 TWO methods of collecting semen**

- Use of an artificial vagina ✓
  - Use of an electro-ejaculator/electrical stimulation ✓
- (2)

**4.6.3 ONE advantage of AI**

- Semen of pure-bred proven/tested bulls can be used ✓
  - Decreases the exchange of sexually transmitted diseases ✓
  - Quick and economical way to improve the herd ✓
  - Semen from bulls from other countries can be used ✓
  - Improves the commercial value of the herd ✓
  - Semen of one bull can be used to AI many cows ✓
  - Semen of superior bulls with desired characteristics can be used long after death/no need to buy/maintain an expensive bull ✓
- (Any 1) (1)

**4.7 Multiple births****4.7.1 Identification of the twins**

**DIAGRAM A** - Dizygotic/fraternal/non-identical twins ✓ (1)

**DIAGRAM C** - Monozygotic/identical twins ✓ (1)

**4.7.2 Diagram representing the twins**

**(a) Of the same sex** - Diagram C ✓ (1)

**(b) Physical appearance differ** - Diagram A ✓ (1)

**4.7.3 The term for the female foetus**

Freemartin ✓ (1)

**4.7.4 The condition of the animal defect**

Sterility ✓ (1)

**4.8 Stages of parturition****4.8.1 Identification of stages****Diagram A** - Ejection of the foetus ✓

(1)

**Diagram B** - Preparatory stage ✓

(1)

**4.8.2 Indication of the stage of parturition**

Ejection/expulsion of the placenta ✓

(1)

**4.8.3 ONE behavioural sign displayed by a cow approaching parturition**

- Cow isolates itself from the herd ✓
- It stops eating ✓
- Cow shows signs of distress and discomfort ✓
- Restlessness, moves in circles, bites or kicks her flank/ walks around ✓
- Makes bellowing noises ✓
- Urinates and defecates often ✓
- Frequent tail raising ✓

(Any 1)

(1)

**[35]****TOTAL SECTION B: 105****GRAND TOTAL: 150**